

Trace Analytical Laboratories, Inc.
2241 Black Creek Road
Muskegon, MI 49444-2673



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888-979-4469 Fax
www.trace-labs.com

June 04, 2025

Patrick Miller
Cadillac, City of
1121 Plett Rd.
Cadillac, MI 49601

RE: Trace Project 25E1239
Client Project NPDES AMR.- Tert. Effluent 5/22/25

Enclosed are your analytical results. The results of this report relate only to the samples listed in the body of this report.

All reports were examined through Trace's validation process to ensure that requirements for quality and completeness were satisfied. All reported analytical results were obtained in accordance with the methods referenced on the reports. Every practical effort was made to meet the reporting limit specifications for this work, however, some results may have raised reporting limits to correct for percent solids.

For clients that require NELAP Accreditation, Trace certifies that these test results meet all requirements of the NELAP Standard, except for those analytes with a "N" notation. These analytes have not been evaluated by NELAP at Trace's discretion and will not be reported unless requested by client.

If you have questions concerning this report, please contact me at 231.773.5998 or by email at tbrewer@trace-labs.com.

Sincerely,

A handwritten signature in black ink that reads "Timothy W. Brewer".

Tim Brewer
Project Manager
Enclosures



TNI EL V1:2016

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SAMPLE SUMMARY

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID	Sample ID	Matrix	Collected By	Date Collected	Date Received
25E1239-01	Tert. Effluent Composite	Aqueous	tl	05/22/25 07:00	05/22/25 13:25
Sample type: Composite					
25E1239-02	Tert. Effluent Grab	Aqueous	tl	05/22/25 09:25	05/22/25 13:25
Sample type: Grab					

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-01 Matrix: Aqueous Date Collected: 05/22/25 07:00
Sample ID: Tert. Effluent Composite Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
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METALS, TOTAL

Analysis Method: EPA 200.7 Rev. 4.4

Batch: T167912

Calcium	76 mg/L	1.0	1	06/02/25	ejb	06/03/25	cma		
Magnesium	18 mg/L	1.0	1	06/02/25	ejb	06/03/25	cma		

Analysis Method: EPA 200.8 Rev. 5.4

Batch: T167912

Antimony	<0.0020 mg/L	0.0020	1	06/02/25	ejb	06/03/25	jma		
Chromium	<0.0050 mg/L	0.0050	1	06/02/25	ejb	06/03/25	jma		
Lead	<0.0030 mg/L	0.0030	1	06/02/25	ejb	06/03/25	jma		
Nickel	<0.0050 mg/L	0.0050	1	06/02/25	ejb	06/03/25	jma		
Selenium	<0.0020 mg/L	0.0020	1	06/02/25	ejb	06/03/25	jma		
Thallium	<0.0020 mg/L	0.0020	1	06/02/25	ejb	06/03/25	jma		
Zinc	0.013 mg/L	0.010	1	06/02/25	ejb	06/03/25	jma		

Analysis Method: SM 2340 B-21

Batch: [CALC]

Hardness as CaCO3	260 mg/L	2.6	1	06/02/25		06/03/25	cma	N	
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SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS

Analysis Method: EPA 625.1

Batch: T167783

N-Nitrosodimethylamine	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Bis(2-chloroethyl)ether	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
2-Chlorophenol	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Phenol	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
1,4-Dichlorobenzene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Bis(2-chloroisopropyl)ether	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
N-Nitrosodi-n-propylamine	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Hexachloroethane	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Nitrobenzene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Isophorone	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
2-Nitrophenol	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
2,4-Dimethylphenol	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-01 Matrix: Aqueous Date Collected: 05/22/25 07:00
Sample ID: Tert. Effluent Composite Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Bis(2-chloroethoxy)methane	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
1,2,4-Trichlorobenzene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
2,4-Dichlorophenol	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Naphthalene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Hexachlorobutadiene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
4-Chloro-3-methylphenol	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Hexachlorocyclopentadiene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
2,4,6-Trichlorophenol	<4.0 ug/L	4.0	1	05/29/25	kbc	06/02/25	jh		
2-Chloronaphthalene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Dimethyl phthalate	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Acenaphthylene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
2,6-Dinitrotoluene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Acenaphthene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
2,4-Dinitrotoluene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
4-Nitrophenol	<20 ug/L	20	1	05/29/25	kbc	06/02/25	jh		
2,4-Dinitrophenol	<20 ug/L	20	1	05/29/25	kbc	06/02/25	jh		
Diethyl phthalate	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Fluorene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
4-Chlorophenyl phenyl ether	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
4,6-Dinitro-2-methylphenol	<20 ug/L	20	1	05/29/25	kbc	06/02/25	jh		
N-Nitrosodiphenylamine	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
1,2-Diphenylhydrazine	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh	N	
4-Bromophenyl phenyl ether	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Hexachlorobenzene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Pentachlorophenol	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Phenanthrene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Anthracene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Di-n-butyl phthalate	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Fluoranthene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Benzidine	<50 ug/L	50	1	05/29/25	kbc	06/02/25	jh	BS01	
Pyrene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Butyl benzyl phthalate	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Benzo (a) anthracene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-01 Matrix: Aqueous Date Collected: 05/22/25 07:00
Sample ID: Tert. Effluent Composite Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
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SEMI-VOLATILE ORGANIC COMPOUNDS BY GC-MS

Chrysene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
3,3'-Dichlorobenzidine	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Bis(2-ethylhexyl)phthalate	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Di-n-octyl phthalate	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Benzo (b) fluoranthene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Benzo (k) fluoranthene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Benzo (a) pyrene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Indeno (1,2,3-cd) pyrene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Dibenz (a,h) anthracene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Benzo (g,h,i) perylene	<5.0 ug/L	5.0	1	05/29/25	kbc	06/02/25	jh		
Surrogates:									
2-Fluorophenol	30 %	20-53	1	05/29/25	kbc	06/02/25	jh		
Phenol-d5	18 %	11-40	1	05/29/25	kbc	06/02/25	jh		
Nitrobenzene-d5	82 %	36-103	1	05/29/25	kbc	06/02/25	jh		
2-Fluorobiphenyl	81 %	36-119	1	05/29/25	kbc	06/02/25	jh	N	
2,4,6-Tribromophenol	73 %	30-105	1	05/29/25	kbc	06/02/25	jh	N	
Terphenyl-d14	60 %	37-109	1	05/29/25	kbc	06/02/25	jh		

Analysis Method: EPA 625.1 SIM

Batch: T167626

Benzidine	<0.10 ug/L	0.10	1	05/27/25	als	05/29/25	jh		
3,3'-Dichlorobenzidine	<1.0 ug/L	1.0	1	05/27/25	als	05/29/25	jh		
Surrogates:									
Nitrobenzene-d5	102 %	38-130	1	05/27/25	als	05/29/25	jh		
2-Fluorobiphenyl	67 %	39-118	1	05/27/25	als	05/29/25	jh	N	
Terphenyl-d14	62 %	14-96	1	05/27/25	als	05/29/25	jh		

PESTICIDES/PCBS

Analysis Method: EPA 612

Batch: T167557

Hexachlorobenzene	<0.0094 ug/L	0.0094	1	05/23/25	kbc	05/23/25	ahr	N	
Hexachlorobutadiene	<0.0094 ug/L	0.0094	1	05/23/25	kbc	05/23/25	ahr	N	
Hexachlorocyclopentadiene	<0.0094 ug/L	0.0094	1	05/23/25	kbc	05/23/25	ahr	N	
Hexachloroethane	<0.0094 ug/L	0.0094	1	05/23/25	kbc	05/23/25	ahr	N	

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-01 Matrix: Aqueous Date Collected: 05/22/25 07:00
Sample ID: Tert. Effluent Composite Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
PESTICIDES/PCBS									
Pentachlorophenol	<1.7 ug/L	1.7	1	05/23/25	kbc	05/23/25	ahr	N	
Surrogates:									
Tetrachloro-m-xylene	61 %	18-105	1	05/23/25	kbc	05/23/25	ahr	N	

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-02 Matrix: Aqueous Date Collected: 05/22/25 09:25
Sample ID: Tert. Effluent Grab Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
Analysis Method: EPA 624.1									
<i>Batch: T167559</i>									
Chloromethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Vinyl chloride	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Bromomethane	<5.0 ug/L	5.0	1	05/23/25	kl	05/23/25	kl		
Chloroethane	<2.0 ug/L	2.0	1	05/23/25	kl	05/23/25	kl		
Acrolein	<10 ug/L	10	1	05/23/25	kl	05/23/25	kl		
1,1-Dichloroethene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Methylene chloride	<5.0 ug/L	5.0	1	05/23/25	kl	05/23/25	kl		
Acrylonitrile	<10 ug/L	10	1	05/23/25	kl	05/23/25	kl		
trans-1,2-Dichloroethene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,1-Dichloroethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Chloroform	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,1,1-Trichloroethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Carbon tetrachloride	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Benzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,2-Dichloroethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Trichloroethene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,2-Dichloropropane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
cis-1,2-Dichloroethene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Bromodichloromethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
2-Chloroethylvinyl ether	<10 ug/L	10	1	05/23/25	kl	05/23/25	kl		
cis-1,3-Dichloropropene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Toluene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
trans-1,3-Dichloropropene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,1,2-Trichloroethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Tetrachloroethene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Dibromochloromethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Chlorobenzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Ethylbenzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Bromoform	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,1,2,2-Tetrachloroethane	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,3-Dichlorobenzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
1,4-Dichlorobenzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		

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ANALYTICAL RESULTS

Trace Project ID: 25E1239
Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

Trace ID: 25E1239-02 Matrix: Aqueous Date Collected: 05/22/25 09:25
Sample ID: Tert. Effluent Grab Date Received: 05/22/25 13:25

PARAMETERS	RESULTS UNITS	RDL	DILUTION	PREPARED	BY	ANALYZED	BY	NOTES	MCL
VOLATILE ORGANIC COMPOUNDS BY GC-MS									
1,2-Dichlorobenzene	<1.0 ug/L	1.0	1	05/23/25	kl	05/23/25	kl		
Surrogates:									
1,2-Dichloroethane-d4	110 %	68-133	1	05/23/25	kl	05/23/25	kl		
Toluene-d8	92 %	75-120	1	05/23/25	kl	05/23/25	kl		
4-Bromofluorobenzene	101 %	69-119	1	05/23/25	kl	05/23/25	kl		
1,2-Dichlorobenzene-d4	101 %	72-127	1	05/23/25	kl	05/23/25	kl		

WET CHEMISTRY

Analysis Method: EPA 420.1
Batch: T167976

Phenolics	<0.010 mg/L	0.010	1	05/30/25	pn	05/30/25	pn
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QUALITY CONTROL RESULTS

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167912

Analysis Description: Calcium, Total

QC Batch Method: EPA 200.2

Analysis Method: EPA 200.7 Rev. 4.4

METHOD BLANK: T167912-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Calcium	mg/L	<1.0	1.0	
Magnesium	mg/L	<1.0	1.0	

LABORATORY CONTROL SAMPLE: T167912-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Calcium	mg/L	16.0	14.6	91	85-115	
Magnesium	mg/L	16.0	15.6	98	85-115	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167912

Analysis Description: Thallium, Total

QC Batch Method: EPA 200.2

Analysis Method: EPA 200.8 Rev. 5.4

METHOD BLANK: T167912-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Chromium	mg/L	<0.00025	0.00025	
Nickel	mg/L	<0.0050	0.0050	
Lead	mg/L	<0.00055	0.00055	
Antimony	mg/L	<0.00025	0.00025	
Selenium	mg/L	<0.00050	0.00050	
Thallium	mg/L	<0.00038	0.00038	
Zinc	mg/L	<0.0012	0.0012	

LABORATORY CONTROL SAMPLE: T167912-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Chromium	mg/L	0.0500	0.0474	95	85-115	
Nickel	mg/L	1.60	1.40	87	85-115	
Lead	mg/L	0.100	0.101	101	85-115	
Antimony	mg/L	0.100	0.106	106	85-115	
Selenium	mg/L	0.100	0.105	105	85-115	

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LABORATORY CONTROL SAMPLE: T167912-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Thallium	mg/L	0.100	0.102	102	85-115	
Zinc	mg/L	1.60	1.47	92	85-115	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: [CALC]

Analysis Description: Hardness (Metals)

QC Batch Method:

Analysis Method: SM 2340 B-21

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167557

Analysis Description: 612 Chlorinated Hydrocarbons

QC Batch Method: EPA 3510C Separatory Funnel
Liquid-Liquid Extr.

Analysis Method: EPA 612

METHOD BLANK: T167557-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Hexachlorobenzene	ug/L	<0.010	0.010	
Hexachlorobutadiene	ug/L	<0.010	0.010	
Hexachlorocyclopentadiene	ug/L	<0.010	0.010	
Hexachloroethane	ug/L	<0.010	0.010	
Pentachlorophenol	ug/L	<1.8	1.8	
Tetrachloro-m-xylene (S)	%	82	18-105	

LABORATORY CONTROL SAMPLE: T167557-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Hexachlorobenzene	ug/L	0.0500	0.0358	72	26-128	
Hexachlorobutadiene	ug/L	0.0500	0.0242	48	26-107	
Hexachlorocyclopentadiene	ug/L	0.0500	0.0259	52	25-105	
Hexachloroethane	ug/L	0.0500	0.0274	55	26-87	
Pentachlorophenol	ug/L	1.00	<1.8	75	25-100	
Tetrachloro-m-xylene (S)	%	0.0600	0.0480	80	18-105	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167783

Analysis Description: TTO Semi-Volatiles

QC Batch Method: EPA 625 Extraction for BNAs

Analysis Method: EPA 625.1

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METHOD BLANK: T167783-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
N-Nitrosodimethylamine	ug/L	<5.0	5.0	
Bis(2-chloroethyl)ether	ug/L	<1.0	1.0	
2-Chlorophenol	ug/L	<2.5	2.5	
Phenol	ug/L	<2.5	2.5	
1,4-Dichlorobenzene	ug/L	<2.5	2.5	
Bis(2-chloroisopropyl)ether	ug/L	<2.5	2.5	
N-Nitrosodi-n-propylamine	ug/L	<2.5	2.5	
Hexachloroethane	ug/L	<2.5	2.5	
Nitrobenzene	ug/L	<1.0	1.0	
Isophorone	ug/L	<2.5	2.5	
2-Nitrophenol	ug/L	<2.5	2.5	
2,4-Dimethylphenol	ug/L	<2.5	2.5	
Bis(2-chloroethoxy)methane	ug/L	<2.5	2.5	
1,2,4-Trichlorobenzene	ug/L	<2.5	2.5	
2,4-Dichlorophenol	ug/L	<2.5	2.5	
Naphthalene	ug/L	<2.5	2.5	
Hexachlorobutadiene	ug/L	<2.5	2.5	
4-Chloro-3-methylphenol	ug/L	<2.5	2.5	
Hexachlorocyclopentadiene	ug/L	<5.0	5.0	
2,4,6-Trichlorophenol	ug/L	<1.8	1.8	
2-Chloronaphthalene	ug/L	<2.5	2.5	
Dimethyl phthalate	ug/L	<2.5	2.5	
Acenaphthylene	ug/L	<2.5	2.5	
2,6-Dinitrotoluene	ug/L	<2.5	2.5	
Acenaphthene	ug/L	<2.5	2.5	
2,4-Dinitrotoluene	ug/L	<2.5	2.5	
4-Nitrophenol	ug/L	<20	20	
2,4-Dinitrophenol	ug/L	<20	20	
Diethyl phthalate	ug/L	<2.5	2.5	
Fluorene	ug/L	<2.5	2.5	
4-Chlorophenyl phenyl ether	ug/L	<2.5	2.5	
4,6-Dinitro-2-methylphenol	ug/L	<5.0	5.0	
N-Nitrosodiphenylamine	ug/L	<2.5	2.5	
1,2-Diphenylhydrazine	ug/L	<2.5	2.5	
4-Bromophenyl phenyl ether	ug/L	<2.5	2.5	
Hexachlorobenzene	ug/L	<2.5	2.5	
Pentachlorophenol	ug/L	<5.0	5.0	
Phenanthrene	ug/L	<1.0	1.0	
Anthracene	ug/L	<2.5	2.5	
Di-n-butyl phthalate	ug/L	<2.5	2.5	
Fluoranthene	ug/L	<1.0	1.0	
Benzidine	ug/L	<50	50	
Pyrene	ug/L	<2.5	2.5	
Butyl benzyl phthalate	ug/L	<2.5	2.5	

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METHOD BLANK: T167783-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzo (a) anthracene	ug/L	<1.0	1.0	
Chrysene	ug/L	<1.0	1.0	
3,3'-Dichlorobenzidine	ug/L	<5.0	5.0	
Bis(2-ethylhexyl)phthalate	ug/L	<2.5	2.5	
Di-n-octyl phthalate	ug/L	<2.5	2.5	
Benzo (b) fluoranthene	ug/L	<1.0	1.0	
Benzo (k) fluoranthene	ug/L	<1.0	1.0	
Benzo (a) pyrene	ug/L	<1.0	1.0	
Indeno (1,2,3-cd) pyrene	ug/L	<1.0	1.0	
Dibenz (a,h) anthracene	ug/L	<1.0	1.0	
Benzo (g,h,i) perylene	ug/L	<1.0	1.0	
2-Fluorophenol (S)	%	20	20-53	
Phenol-d5 (S)	%	11	11-40	
Nitrobenzene-d5 (S)	%	56	36-103	
2-Fluorobiphenyl (S)	%	54	36-119	
2,4,6-Tribromophenol (S)	%	56	30-105	
Terphenyl-d14 (S)	%	69	37-109	

LABORATORY CONTROL SAMPLE: T167783-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Bis(2-chloroethyl)ether	ug/L	100	64.0	64	43-126	
2-Chlorophenol	ug/L	100	69.8	70	41-89	
Phenol	ug/L	100	19.9	20	17-120	
1,4-Dichlorobenzene	ug/L	100	68.5	68	39-78	
Bis(2-chloroisopropyl)ether	ug/L	100	55.4	55	63-139	BS06
N-Nitrosodi-n-propylamine	ug/L	100	77.0	77	14-198	
Hexachloroethane	ug/L	100	70.9	71	55-120	
Nitrobenzene	ug/L	100	76.3	76	54-158	
Isophorone	ug/L	100	73.3	73	47-180	
2-Nitrophenol	ug/L	100	79.9	80	45-167	
2,4-Dimethylphenol	ug/L	100	67.9	68	1-173	
Bis(2-chloroethoxy)methane	ug/L	100	76.9	77	49-165	
1,2,4-Trichlorobenzene	ug/L	100	81.0	81	57-130	
2,4-Dichlorophenol	ug/L	100	77.9	78	54-122	
Naphthalene	ug/L	100	80.5	80	36-120	
Hexachlorobutadiene	ug/L	100	80.7	81	38-120	
4-Chloro-3-methylphenol	ug/L	100	75.5	76	45-93	
Hexachlorocyclopentadiene	ug/L	100	61.2	61	25-105	
2,4,6-Trichlorophenol	ug/L	100	82.2	82	52-129	
2-Chloronaphthalene	ug/L	100	89.7	90	65-120	
Dimethyl phthalate	ug/L	100	83.2	83	1-120	

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LABORATORY CONTROL SAMPLE: T167783-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Acenaphthylene	ug/L	100	89.0	89	54-126	
2,6-Dinitrotoluene	ug/L	100	85.1	85	68-137	
Acenaphthene	ug/L	100	82.3	82	60-132	
2,4-Dinitrotoluene	ug/L	100	84.4	84	48-127	
4-Nitrophenol	ug/L	100	20.7	21	13-129	
2,4-Dinitrophenol	ug/L	100	73.7	74	1-173	
Diethyl phthalate	ug/L	100	83.3	83	1-120	
Fluorene	ug/L	100	88.6	89	70-120	
4-Chlorophenyl phenyl ether	ug/L	100	82.6	83	38-145	
4,6-Dinitro-2-methylphenol	ug/L	100	83.2	83	53-130	
4-Bromophenyl phenyl ether	ug/L	100	86.0	86	65-120	
Hexachlorobenzene	ug/L	100	96.4	96	8-142	
Pentachlorophenol	ug/L	100	51.5	51	38-152	
Phenanthrene	ug/L	100	90.7	91	65-120	
Anthracene	ug/L	100	96.2	96	43-120	
Di-n-butyl phthalate	ug/L	100	94.4	94	8-120	
Fluoranthene	ug/L	100	104	104	43-121	
Benzidine	ug/L	200	<		8-213	BS01
Pyrene	ug/L	100	98.6	99	70-120	
Butyl benzyl phthalate	ug/L	100	109	109	1-140	
Benzo (a) anthracene	ug/L	100	95.7	96	43-133	
Chrysene	ug/L	100	94.5	94	44-140	
3,3'-Dichlorobenzidine	ug/L	200	176	88	8-213	
Bis(2-ethylhexyl)phthalate	ug/L	100	108	108	29-137	
Di-n-octyl phthalate	ug/L	100	99.3	99	19-132	
Benzo (b) fluoranthene	ug/L	100	82.6	83	42-140	
Benzo (k) fluoranthene	ug/L	100	78.4	78	25-146	
Benzo (a) pyrene	ug/L	100	86.3	86	32-148	
Indeno (1,2,3-cd) pyrene	ug/L	100	104	104	1-151	
Dibenz (a,h) anthracene	ug/L	100	99.6	100	1-200	
Benzo (g,h,i) perylene	ug/L	100	118	118	1-195	
2-Fluorophenol (S)	%	100	32.4	32	20-53	
Phenol-d5 (S)	%	100	19.4	19	11-40	
Nitrobenzene-d5 (S)	%	100	87.4	87	36-103	
2-Fluorobiphenyl (S)	%	100	83.2	83	36-119	
2,4,6-Tribromophenol (S)	%	100	99.1	99	30-105	
Terphenyl-d14 (S)	%	100	104	104	37-109	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

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QC Batch: T167626

Analysis Description: 625 Benzidines SIM

QC Batch Method: EPA 625 Extraction for BNAs

Analysis Method: EPA 625.1 SIM

METHOD BLANK: T167626-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Benzidine	ug/L	<0.10	0.10	
3,3'-Dichlorobenzidine	ug/L	<1.0	1.0	
Nitrobenzene-d5 (S)	%	103	38-130	
2-Fluorobiphenyl (S)	%	69	39-118	
Terphenyl-d14 (S)	%	79	14-96	

LABORATORY CONTROL SAMPLE: T167626-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Benzidine	ug/L	2.00	1.16	58	16-104	
3,3'-Dichlorobenzidine	ug/L	2.00	2.73	136	53-133	BS05
Nitrobenzene-d5 (S)	%	1.00	0.985	99	38-130	
2-Fluorobiphenyl (S)	%	1.00	0.671	67	39-118	
Terphenyl-d14 (S)	%	1.00	0.785	79	14-96	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167559

Analysis Description: TTO Volatiles

QC Batch Method: EPA 624 Purge and Trap

Analysis Method: EPA 624.1

METHOD BLANK: T167559-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Chloromethane	ug/L	<1.0	1.0	
Vinyl chloride	ug/L	<1.0	1.0	
Bromomethane	ug/L	<5.0	5.0	
Chloroethane	ug/L	<2.0	2.0	
Acrolein	ug/L	<10	10	
1,1-Dichloroethene	ug/L	<1.0	1.0	
Methylene chloride	ug/L	<5.0	5.0	
Acrylonitrile	ug/L	<10	10	
trans-1,2-Dichloroethene	ug/L	<1.0	1.0	
1,1-Dichloroethane	ug/L	<1.0	1.0	
Chloroform	ug/L	<1.0	1.0	
1,1,1-Trichloroethane	ug/L	<1.0	1.0	
Carbon tetrachloride	ug/L	<1.0	1.0	
Benzene	ug/L	<1.0	1.0	

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METHOD BLANK: T167559-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
1,2-Dichloroethane	ug/L	<1.0	1.0	
Trichloroethene	ug/L	<1.0	1.0	
1,2-Dichloropropane	ug/L	<1.0	1.0	
cis-1,2-Dichloroethene	ug/L	<1.0	1.0	
Bromodichloromethane	ug/L	<1.0	1.0	
2-Chloroethylvinyl ether	ug/L	<10	10	
cis-1,3-Dichloropropene	ug/L	<1.0	1.0	
Toluene	ug/L	<1.0	1.0	
trans-1,3-Dichloropropene	ug/L	<1.0	1.0	
1,1,2-Trichloroethane	ug/L	<1.0	1.0	
Tetrachloroethene	ug/L	<1.0	1.0	
Dibromochloromethane	ug/L	<1.0	1.0	
Chlorobenzene	ug/L	<1.0	1.0	
Ethylbenzene	ug/L	<1.0	1.0	
Bromoform	ug/L	<1.0	1.0	
1,1,2,2-Tetrachloroethane	ug/L	<1.0	1.0	
1,3-Dichlorobenzene	ug/L	<1.0	1.0	
1,4-Dichlorobenzene	ug/L	<1.0	1.0	
1,2-Dichlorobenzene	ug/L	<1.0	1.0	
1,2-Dichloroethane-d4 (S)	%	115	68-133	
Toluene-d8 (S)	%	89	75-120	
4-Bromofluorobenzene (S)	%	101	69-119	
1,2-Dichlorobenzene-d4 (S)	%	102	72-127	

LABORATORY CONTROL SAMPLE: T167559-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Chloromethane	ug/L	50.0	55.3	111	1-205	
Vinyl chloride	ug/L	50.0	65.6	131	5-195	
Bromomethane	ug/L	50.0	52.8	106	15-185	
Chloroethane	ug/L	50.0	56.8	114	40-160	
Acrolein	ug/L	50.0	40.5	81	60-140	
1,1-Dichloroethene	ug/L	50.0	64.1	128	50-150	
Methylene chloride	ug/L	50.0	59.5	119	60-140	
Acrylonitrile	ug/L	50.0	45.2	90	60-140	
trans-1,2-Dichloroethene	ug/L	50.0	63.4	127	70-130	
1,1-Dichloroethane	ug/L	50.0	62.2	124	70-130	
Chloroform	ug/L	50.0	59.5	119	70-135	
1,1,1-Trichloroethane	ug/L	50.0	64.5	129	70-130	
Carbon tetrachloride	ug/L	50.0	67.6	135	70-130	BS05
Benzene	ug/L	50.0	59.7	119	65-135	
1,2-Dichloroethane	ug/L	50.0	63.0	126	70-130	

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LABORATORY CONTROL SAMPLE: T167559-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Trichloroethene	ug/L	50.0	56.6	113	65-135	
1,2-Dichloropropane	ug/L	50.0	58.6	117	35-165	
cis-1,2-Dichloroethene	ug/L	50.0	63.8	128	70-130	
Bromodichloromethane	ug/L	50.0	64.2	128	65-135	
2-Chloroethylvinyl ether	ug/L	50.0	69.5	139	1-225	
cis-1,3-Dichloropropene	ug/L	50.0	59.8	120	25-175	
Toluene	ug/L	50.0	51.5	103	70-130	
trans-1,3-Dichloropropene	ug/L	50.0	56.6	113	50-150	
1,1,2-Trichloroethane	ug/L	50.0	50.5	101	70-130	
Tetrachloroethene	ug/L	50.0	52.6	105	70-130	
Dibromochloromethane	ug/L	50.0	52.5	105	70-135	
Chlorobenzene	ug/L	50.0	49.8	100	65-135	
Ethylbenzene	ug/L	50.0	51.7	103	60-140	
Bromoform	ug/L	50.0	52.5	105	70-130	
1,1,2,2-Tetrachloroethane	ug/L	50.0	51.0	102	60-140	
1,3-Dichlorobenzene	ug/L	50.0	56.2	112	70-130	
1,4-Dichlorobenzene	ug/L	50.0	54.6	109	65-135	
1,2-Dichlorobenzene	ug/L	50.0	53.1	106	65-135	
1,2-Dichloroethane-d4 (S)	%	30.0	33.3	111	68-133	
Toluene-d8 (S)	%	30.0	27.3	91	75-120	
4-Bromofluorobenzene (S)	%	30.0	30.5	102	69-119	
1,2-Dichlorobenzene-d4 (S)	%	30.0	29.5	98	72-127	

Trace Project ID: 25E1239

Client Project ID: NPDES AMR.- Tert. Effluent 5/22/25

QC Batch: T167976

Analysis Description: Phenols, Total

QC Batch Method: EPA 420.1

Analysis Method: EPA 420.1

METHOD BLANK: T167976-BLK1

Parameter	Units	Blank Result	Reporting Limit	Notes
Phenolics	mg/L	<0.010	0.010	

LABORATORY CONTROL SAMPLE: T167976-BS1

Parameter	Units	Spike Conc.	LCS Result	LCS % Rec	% Rec Limit	Notes
Phenolics	mg/L	0.100	0.108	108	73-110	

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AN EXPLANATION OF TERMS AND SYMBOLS WHICH MAY OCCUR IN THIS REPORT

DEFINITIONS

LCS	Laboratory Control Sample
LCSD	Laboratory Control Sample Duplicate
MS	Matrix Spike
MSD	Matrix Spike Duplicate
RPD	Relative Percent Difference
DUP	Matrix Duplicate
RDL	Reporting Detection Limit
MCL	Maximum Contamination Limit
TIC	Tentatively Identified Compound
<, ND or U	Indicates the compound was analyzed for but not detected
*	Indicates a result that exceeds its associated MCL or Surrogate control limits
N	Indicates that the laboratory is not accredited by NELAP for this compound
NA	Indicates that the compound is not available.

NOTE: Samples for volatiles that have been extracted with a water miscible solvent were corrected for the total volume of the solvent/water mixture.
Solid matrices Method Blanks are at 100% solids as such results are the same wet or dry.

DATA QUALIFIERS

Trace ID: 25E1239-01

Analysis: EPA 625.1

Benzidine	Note BS01 : The BS recovery was out of control low. The result and reporting limit for this analyte, in this quality control batch, must be considered estimated.
------------------	---

Trace ID: T167559-BS1

Analysis: EPA 624.1

Carbon tetrachloride	Note BS05 : The BS recovery was out of control high. Because there were no positive results for this analyte in this QC batch, no data require qualification.
-----------------------------	---

Trace ID: T167626-BS1

Analysis: EPA 625.1 SIM

3,3'-Dichlorobenzidine	Note BS05 : The BS recovery was out of control high. Because there were no positive results for this analyte in this QC batch, no data require qualification.
-------------------------------	---

Trace ID: T167626-BSD1

Analysis: EPA 625.1 SIM

3,3'-Dichlorobenzidine	Note BS05 : The BS recovery was out of control high. Because there were no positive results for this analyte in this QC batch, no data require qualification.
-------------------------------	---

Trace ID: T167783-BS1

Analysis: EPA 625.1

Benzidine	Note BS01 : The BS recovery was out of control low. The result and reporting limit for this analyte, in this quality control batch, must be considered estimated.
------------------	---

Bis(2-chloroisopropyl)ether	Note BS06 : Sporadic marginal recovery failure. No data require qualification.
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N-Nitrosodiphenylamine

Note BS06 : Sporadic marginal recovery failure. No data require qualification.

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Page _____ of _____

Report Results To:

Bill To:

Trace Use:

Company Name: City of Cadillac - Utilities - WWTP	PO #:
Report To: Cindy Tomaszewski	Contact Name: Cindy Tomaszewski
Mailing Address: 1121 Plett Rd (for shipping only: UPS, FedEx)	Billing Address (if different): 200 N Lake St (for USPS only)
City, State, Zip Code: Cadillac MI 49601	City, State, Zip Code: Cadillac MI 49601
Office Phone: 231-775-2368	Cell Phone: 231-878-6230
Email Address: lab@cadillac-mi.net, ctomaszewski@cadillac-mi.net	Phone Number: 231-775-0181
	Billing Email Address: ap@cadillac-mi.net

Logged By: GJR	Checked By: MS
Soil Volatiles Preserved (circle if applicable):	
MeOH	Low Level
Sampling Time:	

Turnaround Requirements:

Matrix Key:

☒ Standard, 5-10 Days
☐ 3 Day*
☐ 1 Day*
S = Soil / Solid
W = Water
SL = Sludge
OI = Oil
WI = Wipes
LW = Liquid Waste
A = Air
D = Drinking Water

*Results provided end of business day, requires prior approval.

Project Name: NPDES - Additional Monitoring

Sampled By: T. Laczek

Trace No.	Date Collected	Time Collected	Client Sample ID	Metals Field Filtered (Y / N)	Matrix	Number of Containers	Preservation							Metals, Hardness	SVOC 625, 625 SIMS	Chlorinated Hydrocarbons - 612	VOCs	Total Phenols	Analysis Requested	Remarks	Possible Health Hazards?
							Cool	HCl	HNO ₃	H ₂ SO ₄	NaOH	Other									
1	5/22/2025	0700	Tert. Effluent		W	1														Composite	
1	5/22/2025	0700	Tert. Effluent		W	3														Composite	
1	5/22/2025	0700	Tert. Effluent		W	2	X													Composite	
2	5/22/2025	0725	Tert. Effluent		W	3														Grab	
2	5/22/2025	0725	Tert. Effluent		W	1														Grab	
																					UNKNOWN

Released By	Received By	Date	Time	Released By	Received By	Date	Time
1) T. Laczek	2) H. Hunsicker	5-22-25	0732	3) H. Hunsicker	4) H. Hunsicker	5-22-25	13:25

Please Sign

Check this box if you would not like your samples analyzed if received outside of the conditions outlined in the Trace Sample Acceptance Policy at www.trace-labs.com/downloads.

Form 70-2.1

CERTIFICATE OF ANALYSIS

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888-979-4469 Fax
www.trace-labs.com

25E1239

Cadillac, City of
Project Manager: Tim Brewer

Sample Log In Checklist

Date: 5/22/25	Original Observation	Corrected Temperature	IR-9 (CF: +0.1°C)	IR-12 (CF: 0.0°C)	IR-13 (CF: 0.0°C)	SR1 (CF: -0.4°C)	SR2 (CF: -0.1°C)	Temp Blank	Client Sample
Time: 1640									
Initials: MJ									
Package Description: Cooler	Package Temp °C	0.0	0.0						
	Representative Sample Temp °C	3.0	3.0						

Sample Receipt

Yes No

- ☒ Received on ice or other coolant
☒ Ice still present upon receipt
☐ Custody seals present
☒ Trace Courier ☐ Client Drop-off
☐ Yes ☐ No Custody seals intact (if applicable)
☐ UPS ☐ Fed Ex ☐ US Mail ☐ Other

Sample Condition

Yes No N/A

- ☒ All sample containers arrived unbroken and labeled
☒ Sufficient sample to run requested analyses
☒ Correct chemical preservative added to samples
☐ Samples preserved at Trace
☒ Chemical preservation verified, check EMD pH test strip used (if applicable)
☒ pH 0-2.5 (Lot: HC311850) ☐ pH 11.0-13.0 (Lot: HC022540) ☐ Other
☒ Air bubbles absent from VOAs

Chain of Custody (COC)

Yes No

- ☒ All bottle labels agree with COC
☒ COC filled out properly
☒ COC signed by client

Notes:

Form 70-4.5
Effective 1/30/25

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